Kairav Tupil

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Education

University of Waterloo Waterloo, ON

Bachelor of Mathematics - Double Major: Applied Mathematics with Machine Learning, Statistics

- GPA: 3.5/4.0
- Relevant Courses: Calculus 1–3, Linear Algebra 1 & 2, Stat230 (Probability), CS136 (C), CS231 (Python)
- Recipient of University of Waterloo President's Scholarship

Experience

Machine Learning Developer Sep 2026 – Present

WAT.ai – Offball XI *Waterloo, ON*

- Simulated 30 3v1 episodes in Google Research Football (RL environment) in Python, logging per-step state
- Engineered distance-to-ball feature separating outcomes: all won balls within 0.5 m, all goals beyond 1.7 m
- Quantified sim-to-real gap over 9.7M rows with pandas, NumPy and Matplotlib, finding sim players 2x faster
- Tested and extended World Cup data pipeline (pandas, Parquet) with 153 pytest tests, fixing 5 data bugs

AI Developer Co-op May 2026 – Aug 2026

Compass Group *Mississauga, ON*

- Developed .NET web application with embedded AI assistant, serving 100+ employees self-service support
- Integrated conversational AI agent grounded in 15+ internal documents, automating common employee questions
- Engineered SQL-backed data pipelines integrating structured HR datasets into AI agent knowledge sources
- Automated ETL workflows reducing manual data processing time by 20% organization-wide

Backend Software Developer Oct 2025 – Apr 2026

UW Orbital *Waterloo, ON*

- Engineered FastAPI backend with SQLAlchemy models and middleware supporting a 30+ member team
- Built request/response logging middleware, eliminating blind spots across all API call chains
- Achieved 100% pytest coverage across DB and API layers, eliminating production regressions

Projects

Paper RAG | *Python, LangChain, Chroma, Ollama*

- Built local RAG pipeline over 11 arXiv papers using LangChain, Chroma, and Ollama
- Designed 20-question evaluation harness measuring retrieval accuracy and answer keyword match
- Tuned chunk size and top-k against harness, reaching 19/20 retrieval, 15/20 answers
- Benchmarked cross-encoder reranking over 20 candidates, measuring no gain at added latency

Earshot | *Python, FastAPI, WebSockets, MongoDB, YAMNet*

- Built a FastAPI event switchboard broadcasting real-time sound alerts over WebSockets
- Designed per-user MongoDB Atlas sessions supporting roaming user-taught sounds
- Built a live web dashboard and frontend surfacing classified sounds as they arrived
- Trained a logistic-regression head on frozen YAMNet embeddings, hitting 7/7 alarm recall

Rust Inference Engine | *Rust, Python, PyTorch*

- Built an MLP inference engine in Rust from scratch, using one generalized layer function
- Verified Rust outputs matched PyTorch exactly on 30/30 held-out test samples
- Designed a Python-trains, Rust-serves split, loading exported JSON weights for inference
- Deserialized model weights and test samples into typed Rust structs using serde

Technical Skills & Certifcations

Languages: Python, SQL, HTML/CSS, C, C#, JavaScript, Rust
Frameworks: Flask, FastAPI, Django
Developer Tools: Git, Github, VSCode, Jupyter Notebook, Docker
Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch, OpenCV, LangChain
Certifications: Claude Code 101, Claude Code in Action (Anthropic, July 2026)